BRIDGE TEAM & DECK OFFICER TRAINING

Navigation in Ice

Ice types, voyage preparation, passage planning, watchkeeping and ship handling in ice-covered waters.

IMO Polar Code

SOLAS danger messages

Mariners Handbook NP100

Form BCL-11

COURSE MAP

What we will cover

01 **Ice Types and Formation**
Stages, forms, drift and icebergs

02 **Preparation and Passage Planning**
Briefing, appraisal, planning, execution, monitoring

03 **Extreme Cold and Ice Accretion**
Triggers, consequences and notifications

04 **Preparing the Ship for Ice**
Stores, pipework, ballast and loading

05 **Precautions Under Way and at Anchor**
Bridge measures, damage checks, heavy weather

06 **Responsibilities on Board**
Master, Chief Officer and Chief Engineer

07 **Equipment, Trim and Safety Gear**
Radar limits, stability and protection

08 **Keeping a Keen Watch**
Ice detection and iceberg monitoring

09 **Manoeuvring in Ice**
Entry, engines, ramming and icebreaker work

10 **Reports, Training and Polar Code**
Agencies, risk assessment and compliance

SECTION 01

Ice Types and Formation

01

Ice is an obstacle. It is very strong and commands great respect from mariners who must understand its latent power and strength.

SECTION 1

Stages of sea ice development

Sea ice is classified according to its stage of development, from newly formed crystals to ice that has survived a summer melt.

Stage	Description and thickness
New ice	Recently formed ice composed of ice crystals that are only weakly frozen together, if at all. They have a definite form only while they are afloat.
Nilas	A thin elastic crust of ice up to 10 cm thick, easily bending on waves and swell. Under pressure it grows in a pattern of interlocking fingers, known as finger rafting.
Young ice	Ice in the transition stage between nilas and first-year ice, 10 to 30 cm thick.
First-year ice	Sea ice of not more than one winter's growth, developing from young ice, with a thickness of 30 cm or greater.
Old ice	Sea ice that has survived at least one summer's melt. Its topographic features are generally smoother than first-year ice.

Within each stage there are also sub-types, depending on the internal structure of the ice.

SECTION 1.1

The cycle of sea ice

There is a clear cycle of formation and deformation of sea ice. The process breaks down into four steps.

1. Formation

Ice crystals form in the surface layer and thicken into slush.

2. Growth

The layer thickens and consolidates into a continuous sheet.

3. Deformation

Wind, current and pressure produce ridges and hummocks in the pack.

4. Disintegration

Melting and mechanical break-up return the pack to open water.

SECTION 1.1

The primary forces that move pack ice

Two forces dominate the motion of pack ice: the wind acting on the top surface, and the water acting on the bottom.

Wind stress

At the top surface of the ice

- The wind exerts a force on the surface of the ice pack, causing it to move.
- Ridges and hummocks in the pack present a sail area to the wind.
- Ice with an uneven, rough surface will move faster than smooth ice.
- In the absence of other forces, open pack ice typically moves at 2 per cent of the wind speed.

Water stress

At the bottom of the ice

- Water exerts a drag on the bottom surface of ice being blown across still water, slowing it down.
- The rougher the bottom surface, the greater the drag.
- Where the water is in motion because of a current, it will drag the ice along with it.
- It is essential to consider the presence of sea currents when estimating ice drift.

SECTION 1.1

Three main types of current

Ice drift cannot be estimated from the wind alone. The current regime of the area has to be established first.

Type	Character	Example
Permanent	A constant set and drift that persists throughout the season.	The Labrador Current
Periodic	A predictable current that reverses or varies on a known cycle.	Tidal streams
Temporary	A short-lived current generated by the prevailing weather.	Wind-induced currents

As a general rule, the speed of sea currents gradually decreases with depth.

SECTION 1.1

Estimating ice drift in practice

The relationship between wind speed and ice speed gives a working rule for judging whether ice will move with or against the current.

2%

Speed of open pack ice as a proportion of wind speed

0.5 kn

Example permanent current in a given region

25 kn

Wind needed to move that ice against the current

Depth slows the ice — The greater the depth of the ice, the slower its movement, because current speed decreases with depth.

Why icebergs lag — This explains why icebergs generally move more slowly than the surrounding ice pack.

SECTION 1.2

Common forms of sea ice

Ice takes many forms depending on external conditions and other physical considerations. These are the forms most often met.

Pancake ice

Circular pieces 30 cm to 3 m in diameter and up to 10 cm thick, with raised rims caused by the pieces striking one another.

Brash ice

An accumulation of floating ice made up of fragments not more than 2 m across; the wreckage of other forms of ice.

Ice cake

Any relatively flat piece of ice less than 20 m across.

Floe

Any relatively flat piece of ice 20 m or more across.

Fast ice

Ice which forms and remains fast along the coast.

Ice shelf

Fast ice higher than 2 m above sea level is called an ice shelf.

SECTION 1.2

Why an even sheet of ice seldom forms at once

Except in sheltered waters, an even sheet of ice seldom forms immediately. Two mechanisms explain the sequence.

The slush breaks up

The thickening slush breaks up into separate masses under wind and wave action.

Pancakes form

Those masses take on a characteristic pancake form because the fragments collide with each other.

The sheet consolidates

The slush layer dampens down the waves and, if freezing continues, the pancakes adhere together to form a continuous sheet.

Read the surface

The form of the ice tells the bridge team how it was made and how it will behave. Rough, ridged ice carries more sail area to the wind and more draught below the surface than the smooth sheet it grew from.

What an iceberg is

Freshwater ice

A floating mass of freshwater ice that has broken from the seaward end of a glacier or a polar ice sheet.

Where they are found

Typically in open seas, especially around Greenland and Antarctica.

When they calve

Mostly during spring and summer, when warmer weather increases the rate of calving at the boundaries of the Greenland and Antarctic ice sheets and smaller outlying glaciers.

SECTION 1.3

Iceberg production in numbers

10,000

Icebergs produced each year by the West Greenland glaciers

375

Average number flowing south of Newfoundland into North Atlantic shipping lanes

93%

Share of the world mass of icebergs found surrounding the Antarctic

A hazard to navigation — Those bergs that reach the North Atlantic shipping lanes are a direct hazard to navigation.

Antarctic scale — Antarctic icebergs are not only far more abundant than those of the Arctic, but are of comparatively enormous dimensions.

SECTION 1.3

The size scale of Arctic icebergs

Arctic icebergs vary widely in size, and the smallest are the hardest to detect.

Description	Comparable size
Growler	Roughly the size of a large piano.
Bergy bit	About the size of a small house.
Large berg	Some are the size of a ten-storey building.
Arctic typical	About 45 m tall and 180 m long.
Antarctic	Far more abundant and of comparatively enormous dimensions.

SECTION 1.3

Above and below the waterline

What you can see

- Usually one eighth of an iceberg is above the waterline.
- The part above the water consists of snow and ice which is not very compact.

The cold core

- The ice in the cold core is very compact and therefore relatively heavy.
- It keeps seven eighths of the iceberg under water.
- The temperature in the core is constant, between about -15 °C and -20 °C.

A tumbled berg

- A berg that has tumbled over several times has lost its light snow layers.
- It becomes relatively heavier because of its greater compactness.
- Only about one tenth then rises above the surface.

SECTION 1

Ice terms used on the bridge

The terms below appear throughout this course and in the ice reports the ship will receive.

Finger rafting

Nilas under pressure grows in a pattern of interlocking fingers.

Ridging

Deformation of the pack that raises ridges; ridged ice may be far too deep and could severely damage the hull.

Hummocking

Deformation producing hummocks, which together with ridges present a sail area to the wind.

Ice blink

A luminous yellow haze on the horizon in blue skies, or a white glare on the clouds under overcast, in the direction of the ice.

Calving

The separation of an iceberg from a glacier or ice sheet.

Ice edge

The definite beginning of the pack, and the point of entry to it.

THE FIRST PRINCIPLE

Seven eighths of every iceberg is out of sight

Icebergs can appear small above the surface but be very large beneath it. They can be concealed in fog, or blend in with a grey sky and sea. Nothing you can see tells you where the underwater spurs are.

One eighth above water

Seven eighths below

One tenth if tumbled

SECTION 02

02

Preparation and Passage Planning

Meticulous voyage planning is necessary for ice navigation, and any instructions issued by the classification society must be complied with.

Approval, checklist and class limits

Ice navigation is not a decision taken on board alone. Three conditions apply before the vessel goes anywhere near ice.

Company approval first

Navigation in ice may only be conducted after consultation with, and approval from, the Company Operations Department.

Complete Form BCL-11

The checklist Form No. BCL-11 shall be completed prior to, during and after entry into an area of ice, or whilst navigating close to such an area.

Strengthened and classified only

Only vessels which are properly strengthened and classified for ice navigation should contemplate such operation, subject to the limits of the classification.

Before anything else

If it is established that the vessel will be entering an ice zone, the Master needs to call his officers together for a voyage briefing. The briefing is the point at which the whole plan is agreed and understood.

The Master’s voyage briefing

The briefing given to the officers before entering an ice zone should include the following.

- Risk assessment for the intended operation
- Time of year and the expected conditions
- State of the hull, machinery and equipment
- The vessel’s stability and her ability to control icing up
- Ice experience of the officers and crew
- Area of operation and access to icebreakers
- Laws that can differ between territories
- Voyage duration

Experience is part of the plan

The Master needs to know the ice experience of the individual officers and how it relates to his own. The briefing is where that is established, not at the ice edge.

SECTION 2.1

Where current ice information comes from

All available and current information on ice, including icebergs and pack ice, should be gathered before entering any waters where ice may be encountered.

Admiralty List of Radio Signals

Broadcast schedules and stations for ice information.

US Coast Guard

International Ice Patrol reports.

UK Met Office

Meteorological and ice information.

Baltic Sea Ice Code

Baltic ice code reports.

Canadian Coast Guard

Ice Advisory Service broadcasts.

SECTION 2.1

What to review before entering ice waters

Particular regard should be given to the following before entering any water where ice may be encountered.

Draught and trim
The loaded condition must suit ice navigation, not just the passage.

Propeller immersion
The propeller must be kept below the ice level throughout.

Use of rudder
How the rudder will be used, and the limits placed on its movement.

Propulsion and auxiliary machinery
Operational availability of the main propulsion and the auxiliary machinery.

Mooring equipment and anchors
Condition and availability of mooring equipment, including the anchors.

Class instructions
Any instructions issued by the classification society must be complied with.

Radar limitations and the reference library

Radar will not see everything

- Masters should be aware of the limitations of radar in detecting ice.
- The consequence is a need to increase lookouts, not a need to trust the screen.

Mariners Handbook NP100

- Reference should be made to the relevant chapter of the Mariners Handbook, NP100.
- Meticulous voyage planning is necessary for ice navigation.
- Company passage planning requirements are followed at the same time.

SECTION 2.1.1

Four principles of any passage plan

A passage plan into ice-infested waters is no different from any other and is constructed along the same four principles.

1. Appraisal

Gather all information affecting the proposed passage, with additional ice considerations.

2. Planning

Mark the ice limits and build in contingencies and special ice standing orders.

3. Execution

Adjust course and speed as conditions require, with the Master's approval.

4. Monitoring

Fix meticulously and monitor weather reports and ice positions throughout.

SECTION 2.1.1

What makes an ice passage plan different

Detailed passage planning is the key to completing a successful voyage through ice-covered waters.

Seasonal routeing trends

More attention is paid to seasonal and weather routeing trends than on an open-water passage.

Communication emphasis

A greater emphasis is placed on establishing and maintaining communication.

Limits on the chart

Pack ice and iceberg limits are highlighted on the navigational chart.

Reported ice positions

Those limits are enhanced by reported ice positions issued by the ice patrol or similar reporting organisations.

SECTION 2.1.1

Appraisal — the additional considerations

Gathering all information affecting the proposed passage is normal practice for any plan. In ice conditions there are additional considerations.

The geography can change

The geography of the route could well be affected by ice growth or ice movement that threatens the safety of the ship.

Ice chartlets and ice reports

Standard information should be enhanced by facsimile ice chartlets and ice reports.

Reporting organisations

Issued by organisations such as the USCG, the International Ice Patrol, Punta Arenas Radio, Kiel Harbour Radio and Antarctic weather and research stations.

The designated ice officer

The Meteorological Office has a designated ice officer who can be consulted for the intended area.

Appraisal — limits, corrections and signals

- The standard ice limits for both pack ice and icebergs should be confirmed.
- Any seasonal changes to those limits should be noted.
- Weekly notices for the latest chart corrections should be scrutinised.
- Charts must be brought up to date prior to starting the voyage.
- Where icebreaker contact is anticipated, the International Code of Signals is still applicable, although it is seldom used in practice.

Signals still count

All officers on the bridge must be thoroughly familiar with icebreaker signals as shown in the International Code of Signals, even though radio communication is used in practice.

SECTION 2.1.1.2

Planning — marking the ice limits

Where great circle or rhumb line tracks intersect with known ice limits in the ice season, those limits should be marked on the charts.

Mark the intersection
Marking the ice limits shows exactly where the intended track meets the ice.

Upgrade watchkeeping early
This allows the Master to upgrade watchkeeping duties to combat any threat from ice before it is encountered.

Voyage duration
Voyage duration must be considered during the planning stage.

Special ice standing orders
Special ice standing orders should be followed in addition to all routine inclusions in the passage plan.

Contingency for heavy ice
Contingency planning for heavy ice should be included if alternative choices are available.

SECTION 2.1.1.2

Planning — position fixing and radar confusion

Position fixing methods for use with the plan should be examined in detail to ensure primary and secondary systems are possible in all conditions.

What the plan must provide

- Primary and secondary fixing systems that work in the expected conditions.
- Alternatives for when ice obscures the horizon and any navigation marks.

How radar targets deceive

- Headlands can appear to be extended by fast ice.
- Island targets can be mistakenly identified as icebergs.
- Islands can blend with the mainland when connected by fast ice.

PLANNING PRINCIPLE

Planned berth to berth, but never fixed

All passage plans are designed to cover the passage from berth to berth but, by their very nature, they must remain flexible to the changing environment. This is especially relevant in conditions of moving ice formations.

Berth to berth

Always flexible

Master approves every change

SECTION 2.1.1.3

Execution — entering the ice

Deviations in course or speed inside ice limits are not unusual, but every change is controlled and recorded.

DO

- Log every change of course and speed inside the ice limits.
- Change the revised waypoint positions to suit the ongoing circumstances.
- Confirm and obtain the Master’s approval for every change.
- Notify the Master of the ship’s approach to the ice edge before entry.
- Make the actual entry at reduced speed, always.
- Support the watch officer in adjusting speed when necessary, to the point of operating full astern.

DON'T

- Do not enter ice without the Master having been notified of the approach to the ice edge.
- Do not assume the hull will escape damage; ice damage must be anticipated in pack ice.
- Do not combine high speed with heavy ice; the two are totally incompatible.
- Do not alter waypoints without the Master’s confirmation and approval.
- Do not hesitate to reduce speed while waiting for permission.

SECTION 2.1.1.4

Monitoring — position and ice

In icy waters, position monitoring must be meticulous, and this is more difficult to achieve than in open water conditions.

GPS helps, but is not enough
The use of Global Positioning Systems has helped considerably, but primary and secondary systems should also be employed throughout the voyage.

Tight position fixing
Where ice surfaces dominate, it is essential that tight position fixing is followed.

Corroborate with under-keel clearance
Fixes must be corroborated by under-keel clearance at all times.

Slowing down is an advantage
There is a benefit in the passage slowing down in ice: it provides more time to determine your position.

Weather and ice positions
Monitoring of weather reports and ice positions is an essential element of ice navigation.

SECTION 2.1.1.5

Weather routeing — the variable factors

Many ocean passage plans direct ships through ice-affected waters, and routeing organisations need to investigate three groups of variable factors.

Geographic criteria

The geography of the route and the ice limits that cross it.

Ship specification and capability

What the ship is built and classed to do, and what she can actually sustain.

Seasonal meteorological patterns

The expected weather and ice pattern for the season of the voyage.

Planning near ice limits — the ship and the charter

When planning a route near ice limits in either hemisphere, the navigation officer should consider the following.

- The recommended route must consider the safety of the ship, passengers, crew and cargo, and must not stand the ship into danger.
- The expected dangers from ice in season, fog and storms, particularly in the North Atlantic and the Southern Ocean regions.
- The classification of the ship with regard to its quality of ice strengthening.
- Desired speed, maximum speed and charter party specified speed for the voyage.
- Charter party clauses about limitations of latitude.
- Special needs of the ship concerning cargo requirements or shipboard precautions.

Commercial pressure is not a reason

Charter party speed and latitude clauses are inputs to the plan, not overrides. The recommended route must not stand the ship into danger.

SECTION 2.1.1.5

Planning near ice limits — passage and weather

Projected weather

Projected weather patterns for the period of the voyage.

Hazards and clearance

Proximity of additional navigation hazards and adequate under-keel clearance throughout the passage.

Endurance and bunkers

Endurance and bunker capacity of the ship for the chosen route.

Ocean Passages of the World

Recommendations from Ocean Passages of the World.

Prognosis charts

Prognosis charts for the ship in various sea conditions and wave heights.

Master's experience

The Master's experience and preference in the choice of route.

Seasonal ice limits

Seasonal ice limits and iceberg-bearing currents.

Fixing aids available

Position fixing and monitoring aids actually available on the route.

SECTION 2.1.1.5

Position fixing limitations in pack ice

Several routine navigational methods become unreliable or unavailable once the ship is in pack ice.

Speed is irregular
Speed is irregular and variable in pack ice conditions.

Systems may be unavailable
Primary and secondary position monitoring systems may not always be available in ice conditions.

Visual fixing impractical
Visual fixing options may be impractical because of the environment.

Dead reckoning may fail
Dead reckoning requires compass direction and logged distance; compass unreliability and unavailable log data may make it impossible.

Equipment maintenance
Navigation equipment may experience maintenance problems in extreme weather conditions and low temperatures.

SECTION 03

Extreme Cold and Ice Accretion

03

The onset of ice accretion can be sudden and dangerous. Driving sea spray adheres to the ship's structure and forms ice which builds up rapidly.

SECTION 2.1.2

What counts as extreme conditions

Any one of the following conditions is considered as extreme conditions. When they are encountered or expected, Masters apply the instructions and procedures for ice.

Trigger	Threshold or circumstance
Air temperature	Below 2 °C (35 °F).
Sea water temperature	Below 0 °C (32 °F).
Floating ice	Floating ice is reported in the vicinity.
Reports from other vessels	Reports received from vessels in the same area advising sighting of ice, or advising winds causing ice accumulation.
Wind-driven accumulation	Winds of any speed causing any ice accumulation on the vessel.
Precipitation	Snow or freezing rain.

When the above conditions are encountered or expected to be encountered, Masters are instructed to apply the instructions and procedures contained within the procedure.

SECTION 2.1.2

How ice accretion happens, and what to record

The onset can be sudden

The onset of ice accretion can be sudden and dangerous, with little warning to the bridge team.

Driving spray builds ice

Driving sea spray adheres to the ship's structure and forms ice which builds up rapidly.

Ice on the sea surface

If there is ice formation on the sea surface this usually restricts the sea spray; however, accretion is still possible where there is high air moisture content.

Compile the record

All messages, bell book entries, logbook entries and other received information are to be carefully compiled for future claims, investigations of conduct or other matters.

Records protect the ship

Masters are advised that the record made during the passage is the evidence relied on afterwards. It must be compiled while the vessel is in ice and extreme cold conditions, not reconstructed later.

SECTION 2.1.2

The consequences of ice accretion

Ice building up on the structure changes the ship’s condition and disables systems the crew relies on.

Change in trim

Change in trim, generally by the head.

Reduction in stability

Loss of stability as weight is added high up in the ship.

Blocked air vents

Blockage of air vents, including ballast tank air vents.

Frozen pipelines

Freezing of pipelines throughout the exposed systems.

Hydraulic oil viscosity

Increase in the viscosity of hydraulic oil in the systems.

Fracturing castings

Fracturing of exposed castings made from cast iron.

Icing of bridge windows

Loss of visibility from the conning position as bridge windows ice over.

Frozen intakes

Cooling water intakes may also become frozen.

SECTION 2.1.2

What ice on the sea surface means for the ship

Forcing causes damage

- Ice formation on the sea surface severely restricts navigation.
- Forcing the ship through ice can cause damage to the hull and the propeller.
- Damage is also caused if attempts are made to go astern.

You may need help

- The assistance of an icebreaker could be required.
- The vessel may have to join an ice convoy.

Therefore

- The decision to enter an ice field has to be carefully considered.
- It is a decision for the Master with Company approval, not a routine navigational choice.

THE DECISION TO ENTER

Entering an ice field is never a routine decision

Ice is an obstacle to the progress of any vessel, and is dangerous to a vessel not specially equipped or constructed for ice navigation. The most serious danger to a ship in ice is pressure, which may result in damage to the hull or the ship's bottom.

Company approval required

Class limits apply

Pressure is the greatest danger

SECTION 2.1.3

Notifications — who must be told, and when

Reporting of conditions is required by SOLAS as danger messages when encountering any ice, gale force weather and ice accumulation.

SOLAS danger messages

Any ice, gale force weather or ice accumulation encountered is reported as a danger message under SOLAS.

On receipt of voyage orders

The Charterer and the Company or Owner are notified immediately when voyage orders instruct the vessel to proceed to areas where ice or extreme cold conditions may be encountered or expected.

On encountering the conditions

They are notified immediately when any of the ice or extreme cold conditions listed is encountered.

Immediately means immediately

Notification is not held until the end of the watch or the next scheduled report.

What the Master reports to the Company

- The exact status of equipment and stores on board to deal with ice or extreme cold conditions.
- Regular advice on the conditions actually being encountered.
- Any reports of injuries, damages received or delays experienced, made promptly on the appropriate form.
- Logbook entries and photographs attached to those reports.
- A request that the Company advise whether the vessel is suitably covered by insurance for the intended voyage.
- Advice of the expected extra bunker consumption due to cold weather operations.

Attach the evidence

Damage and delay reports are made on the appropriate form with logbook entries and photographs attached. A report without evidence is of little use in a later claim.

SECTION 04

Preparing the Ship for Ice

04

The following are the minimum requirements for vessels proceeding into ice or extreme cold conditions.

SECTION 2.1.4

Minimum requirements before proceeding into ice

Complete the risk assessment

The risk assessment regarding precautions, preparations and necessary controls in ice must be done in all respects.

Store and equip the ship

The vessel shall be stored and equipped with antifreeze agents, deck non-slip material, temporary repair material and fuel oil additives.

Advise the Chief Engineer early

The Chief Engineer must be advised well before expected temperature reductions, and then kept advised of the actual temperatures.

Pilot, tug or icebreaker standby

Consideration is to be given to keeping a pilot on board and a standby tug or icebreaker at all times, even after berthing and during cargo operations, if ice conditions are considered dangerous.

SECTION 2.1.4

Stores and consumables to carry

The vessel shall store and equip with the following before proceeding into ice or extreme cold conditions.

Category	Items
Antifreeze agents	Methanol, ethanol, glycol, engine cooling antifreeze and de-icing salt.
Deck non-slip	Sand for treating working decks and access routes.
Temporary repair material	Plastic steel, Devcon, cordo-bond and epoxy glues.
Fuel oil additives	Non-solidifying additives for the fuel oil system.
De-icing compounds	An adequate supply of one of the proprietary commercial de-icing compounds.

SECTION 2.1.4

Water systems and pipework in freezing conditions

Care must be taken to avoid damage, and the associated danger to personnel, caused by freezing water in deck lines, pipe tunnels and void space pipelines and fittings.

DO

- Keep water systems circulating if they cannot be stopped and drained.
- Blow all unused piping dry with air after draining.
- Prepare steam hoses or hot water hoses for defrosting.
- Pay particular attention to reducing damage in pipe tunnels, deck trunks and void spaces.
- Keep sea water strainers and filters clean.

DON'T

- Do not leave water standing in deck lines, pipe tunnels or void space pipework.
- Do not assume a drained line is dry; it must be blown through with air.
- Do not leave defrosting arrangements to be found when they are needed.
- Do not send personnel to work on frozen fittings without assessing the danger first.

SECTION 2.1.4

Ballast tank precautions

For ballast tanks, consideration shall be given to the following.

- Removing the forward-most vent head to stop it freezing over.
- No ballasting is to be carried out without first checking that the venting system is operational.
- Pressure, vacuum and high velocity relief valves should be checked frequently during cold weather to ensure frozen condensate is not affecting their correct functioning.
- All tank venting arrangements are to be cleared to ensure continuous venting of tanks.
- Keeping ballast tanks slack to avoid upper tank pipes freezing over; consideration should be given to keeping tanks at 90 per cent of capacity where possible.
- The vessel should maintain the deepest drafts possible by loading cargo prior to discharging ballast.

Why 90 per cent

Tanks should be no more than 90 per cent full so that there is room to accommodate expansion if the contents freeze. A full tank with a frozen vent has nowhere to go.

SECTION 2.1.4

Gathering information for appraisal

All information from the following sources is to be gathered to assist with proper situation appraisal and proper route planning.

Ice reporting services

Ice charts from the recognised ice reporting services for the area.

Commercial parties

Agents, owner, Company, charterers and shippers.

Meteorological offices

Forecasts and ice information from the meteorological offices covering the route.

Publications

BIMCO, pilot books, Guide to Port Entry and the Mariners Handbook.

NAVTEX ice reports

Ensure the vessel's NAVTEX has the ice report receiving function activated.

Weather routeing services

Routeing should as far as practicable avoid or minimise time in ice or extremely cold conditions; the use of weather routeing services is to be considered.

SECTION 2.1.4

Three numbers that govern speed in ice

The vessel is not to force ice. Forcing ice has a definition, and so does the speed at which packed ice is navigated.

50%

Reduction of normal speed that counts as forcing ice

5 kn

Generally the maximum speed for navigating through packed ice

3 ft

Maximum trim, to keep ice sliding under the vessel

Forcing ice defined — Forcing ice is to be considered breaking close ice, or having the ship's normal speed reduced by half because of trying to pass through ice.

Why trim matters — Less than three feet of trim keeps ice sliding under the vessel from reaching the sea suction and the propeller below the ice level.

Reduce to minimum — Reduce speed to minimum, with the main engine on diesel fuel if necessary.

SECTION 2.1.4

Reference publications for ice work

Prior to and during navigation in ice or extreme cold conditions, reference is to be made to the following, and the Master is instructed to use the information to best advantage.

Ice Handbook — BIMCO

Reference guidance on ice procedures for commercial shipping.

Mariners Handbook

The relevant chapter of NP100 on ice and ice navigation.

ISGOTT

ISGOTT guidance on ice procedures for tankers.

What to look for

Particular guidance is to be sought to help ensure that the hull and the propeller are not damaged.

Never underestimate the ice

Never underestimate the hardness of ice. It can vary in thickness, the floes can be of different sizes and therefore of different strengths, and it can also be moving in a current.

SECTION 2.1.4

Routeing, information and forcing ice

Routeing should, as far as practicable, avoid or minimise time in ice or extremely cold conditions.

DO

- Gather all information from ice reporting services, agents, owner, Company, charterers, shippers and meteorological offices.
- Consult BIMCO, pilot books, the Guide to Port Entry and the Mariners Handbook.
- Ensure the vessel's NAVTEX has the ice report receiving function activated.
- Consider the use of weather routeing services.
- Plan the route to avoid or minimise time in ice or extremely cold conditions.

DON'T

- The vessel is not to force ice.
- Do not break close ice in order to make progress.
- Do not allow the ship's normal speed to be reduced by half in an attempt to pass through ice; that is forcing ice.
- Do not plan a route through ice where a practicable alternative exists.

SECTION 2.1.4

Thermal shock during loading

Due to rapidly changing temperature, the highest stress on the hull structure occurs during loading operations when hot cargo is loaded into cold tanks.

Parameter	Value or required action
Air temperature	+6 °C or below.
Sea water temperature	-2 °C or below.
Cargo to tank temperature difference	55 °C.
Loading rate	Keep the loading rate low, with cross tank loading, to prevent thermal shock on the hull structure.
Notification	The Company is to be informed before commencement of the loading operation, and the loading plan must be revised as slow and cross loading.

The three conditions above are considered together when deciding whether the loading plan must be revised.

SECTION 05

Precautions Under Way and at Anchor

05

What the bridge team does once the ship is in ice, how damage is checked for, and how the ship is handled in adverse weather.

SECTION 2.1.4

Bridge measures when navigating in ice

The following should be taken into account if in ice.

- Pay more attention when sailing during darkness through ice-packed zones, using the searchlight.
- Extra watches are to be set when encountering ice, and consideration made to putting a lookout forward.
- Try to be in contact with the weather and ice control station, the pilot station and the agents.
- Constant monitoring of the radar screen to observe ice formations such as paths in advance.
- Reduce speed to minimum, with the main engine on diesel fuel if necessary.
- Generally try to navigate through packed ice at speeds of less than 5 knots.

Searchlights at night

At night in light drift ice all searchlights must be used, with the helmsman in place ready to take evasive action. Searchlights should be of a narrow beam type and fitted with an adequate means of de-icing.

SECTION 2.1.4

Engines and rudder when the ship is in heavy ice

If the vessel becomes ice bound, or is under way but restricted in heavy ice, main engine operations should be kept to a minimum.

DO

- Use minimum power ahead to keep steerage way.
- Inspect the propeller for possible damage in this situation, if it is possible to do so.
- If forced to stop, put the rudder amidships and keep the engine turning slowly ahead.
- Keep the engine running and the propeller turning where possible, to prevent ice forming at the stern.
- Keep sea water strainers and filters clean.

DON'T

- Do not make astern movements unless they are unavoidable.
- Do not use hard over rudder except in an emergency.
- Do not anchor in ice conditions.
- Do not force ice; breaking close ice or halving the ship’s normal speed to get through is forcing.

SECTION 2.1.4

Channels, drift and compass checks

Drift in narrow channels

Be aware of the drifting effect of ice when passing through narrow channels, especially when combined with the effect of wind and currents.

In an icebreaker's channel

If in a channel made by an icebreaker, follow strictly the pilot's or icebreaker's advice and commands.

Check speed and course often

Check the speeds and courses frequently as conditions change.

The magnetic compass near the poles

Remember that the magnetic compass is of little value near the poles.

Check the gyro by azimuth

Check the gyro compass frequently by azimuth bearings.

SECTION 2.1.4

Damage checks and ballast condition

- To prepare for any flooding, all watertight doors as appropriate are to be shut.
- Check for any hull plating damage that may have occurred after transit through heavy ice, by careful inspection of all empty spaces.
- Inspections should be made in very cold weather after hard impacts with docks or fenders.
- At the end of sea watches, regular checking of the preparations and measures taken is to be carried out.
- Ballast condition to maximise propeller immersion, sea chest intake immersion and forward draft is to be considered.
- The condition is to be reported to the Company or Owner and the Charterer with details of any operations carried out or expected on board to counteract ice or heavy weather.
- The vessel should have less than three feet of trim, to keep ice sliding under the vessel from reaching the sea suction and propeller below the ice level.

Look inside, not just outside

Hull plating damage from heavy ice is found by careful inspection of all empty spaces. External appearance after a transit tells you very little.

Daily instructions and rapid temperature change

Written instructions daily

- The Master's written instructions are to be made at least daily.
- They refer to the climatic conditions and the precautions to be carried out.

Rapid temperature change

- Under certain circumstances rapid changes of temperature may be experienced.
- It is the responsibility of the Master and the duty deck officers to advise the Chief Engineer when this occurs.
- This allows proper precautions to be taken to prevent freezing and damage to machinery.

Prepare ship and crew

- The Master must take all necessary precautions to prepare ship and crew for operation in unduly cold weather.
- All departments must be familiarised with the precautions to be taken to avoid damage, illness and injury from the expected cold conditions.

SECTION 2.1.4

Looking after the crew in extreme cold

Hypothermia and frostbite

First aid carers on board are to be familiarised with the hypothermia and frostbite sections of the Ship Captains Medical Advice Book and the Mariners Handbook.

Accommodation heating

Crew accommodation heating is to be tested and checked for operational readiness.

Survival clothing

The crew must have the appropriate working and survival clothing for the conditions. A poorly equipped sailor will not last long and cannot carry out his duties or survive at emergency or abandon ship stations.

The cold to expect

Ambient temperatures can be -20 °C, with a wind chill equivalent to -30 °C.

SECTION 2.1.5

Adverse weather when under way

Additional securing

The Master is to ensure that additional securing arrangements are made in anticipation of adverse weather.

Check every three days

The securing arrangements should be checked every three days during prolonged periods of heavy weather.

Alter course if needed

This may mean altering course to allow safe access to the forecastle.

Access is part of the plan

A check that cannot be carried out safely will not be carried out at all. Where access to the forecastle is unsafe on the present heading, the course is altered to make the check possible.

Adverse weather at anchor

- If the Master or officers are in any doubt as to the security or safety of the vessel, the engines must be immediately brought to readiness.
- The Master should assess the situation and, if thought necessary, weigh anchor.
- This decision should not be left too late, so that additional weight is not put to bear on the cable and windlass when heaving.
- The stress on the cable can be eased by using the engine, or by using the rudder to reduce yawing, especially in a current.
- Frequent weather forecasts must be obtained and evaluated when the vessel is anchored.
- Assessment should be made of the wind direction, whether it is a lee shore, and what the holding ground is like.
- Considerable dynamic stress on the anchor system may be induced by yawing of the ship under the influence of wind and tide.
- Yawing can be moderated by lowering a second anchor on to the sea bed.

Lowering the second anchor

When the second anchor is being made ready, the anchor and the shank are to be lowered inside the water to dampen the effect of the anchor banging against the ship’s side due to rolling.

SECTION 06

Responsibilities on Board

06

The Master takes overall responsibility, but the Chief Officer and Chief Engineer each carry a defined part of the preparation for ice.

SECTION 2.2

The Master — overall responsibility

Overall responsibility

The Master takes overall responsibility and should have received detailed instructions on the particular voyage.

Know the experience on board

He needs to know the ice experience of the individual officers and how this relates to his own experience.

Make the dangers explicit

He must ensure that his officers are completely aware of the dangers, and how these could affect their individual tasks.

Provide the information

He must provide all relevant meteorological information to the bridge team.

If the Master has ice experience he must

- Be aware of the ice class limitations of the vessel, with regard to items such as the ice condition and the vessel's draft.
- Note that local laws and regulations for ship ice classification in the Baltic differ in the territorial waters of Sweden, Finland and Russia.
- Be aware of the availability of local ice pilots.
- Order a check on the ship's external systems to make sure all are prepared for ice and snow conditions.
- Ensure that the vessel has an adequate GM for probable icing up conditions and, where appropriate, for any icebreaking that is required.
- Obtain information on the availability of icebreakers, and understand how to captain his ship under the direction of the icebreaker captain.
- Include the ice precautions in the Master and pilot exchange where applicable.

If experience is limited

If the Master's own experience is limited, it is his own responsibility to take all necessary steps to acquire the relevant knowledge and skills needed to remain in full command.

Ice classification and local rules

Ice class limitations

The vessel's ice class sets limits with regard to the ice condition and the vessel's draft. Operation must stay inside them.

Baltic rules differ

Local laws and regulations for ship ice classification in the Baltic differ in the territorial waters of Sweden, Finland and Russia.

Local ice pilots

The Master must be aware of the availability of local ice pilots for the area of operation.

SECTION 2.2

The Chief Officer

Read it before the ice

- The Chief Officer must make sure he has read all the relevant documentation.
- This includes the International Maritime Organization recommendations for navigating in ice.

Ballast, trim and systems

- He needs to know how to ballast the ship for ice and maintain her trim.
- He must ensure the many ship's systems exposed to temperature change are protected so that they continue functioning.

Know his officers

- He also needs to know the ice experience of his fellow officers.
- He must be assured that they understand their responsibilities.

Chief Officer — winter precautions and towing

Consideration should also be given to the following.

Manufacturer’s winter precautions

Equipment manufacturers’ instructions with regard to winter precautions, for example hydraulic oil for cranes and mooring equipment.

Deck IG water seals

The deck inert gas water seals need specific attention in freezing conditions.

Accepting a tow

How the vessel can be prepared to accept a tow by an icebreaker.

SECTION 2.2

The Chief Engineer

Engines and systems ready

The Chief Engineer's job is to make sure the ship's engines and systems are prepared for ice conditions and will keep functioning as conditions worsen.

Know the vulnerable parts

He needs to be aware of the dangers ice poses to specific functions such as sea chests, bow thrusters, propellers and rudders.

Internal water circulation

He needs to make sure the internal water circulation is functioning.

Carry out the checks

He should make sure that the necessary checks, such as those to the ship's external systems, are carried out.

Sea chest efficiency

He must ensure that the cooling water sea chests are working at optimum efficiency.

Chief Engineer — oils, power and manpower

- Are the correct lubrication and fuel oils available, so that they will not freeze or crystallise on board?
- He needs to know what levels of ice are expected, and whether the vessel will be required to force its way through the ice.
- He requires that knowledge so that he can be sure the engine has sufficient power to maintain manoeuvrability.
- Even though most ships are now on bridge control of the main engine, the Chief Engineer still needs to be kept fully informed.
- Being informed allows him to allocate his limited manpower accordingly.
- He must be kept informed of the moment of entry into the ice, so that he can respond to demands for extra power immediately.

Keep the engine room in the picture

Regular communications with the engine room should be maintained at all times, so that the Chief Engineer and his staff are aware of the prevailing conditions and are ready to increase and reduce power when necessary.

SECTION 2.2

Who does what in ice — summary

Role	Primary responsibility in ice
Master	Overall responsibility; briefing, reporting to authorities and the Company, ice class limits, GM, icebreaker liaison and daily written instructions.
Chief Officer	Documentation and IMO recommendations, ballast and trim for ice, protection of exposed systems, preparation to accept a tow.
Chief Engineer	Engines and systems ready and kept running, sea chests, bow thrusters, propeller and rudder, oils, power for manoeuvring and manpower.
Chief Mate	Usually assists the Master in conning the vessel from the bridge when navigating in ice.
Officer of the Watch	Identification of ice types and formations, lookout arrangements, radar monitoring and continuous position plotting.

SECTION 07

Equipment, Trim and Safety Gear

What radar can and cannot detect, the trim and stability the ship must hold, and the protection the crew must be given.

07

SECTION 2.3

Radars — maintenance and running

Well maintained

Have the radars been well maintained? The check is made before the ice is reached, not after.

Run them constantly

They will need to run constantly to avoid freezing.

Study the returns in clear weather

Every opportunity should be taken in clear weather to study the radar returns from all the different types and concentrations of sea ice, to assist watchkeepers in their assessment in reduced visibility.

SECTION 2.3

Radar detection ranges in calm conditions

In calm or relatively calm conditions radar is reliable for detection at the following ranges.

Target	Reliable detection range
Icebergs	Up to a range of between 12 and 15 miles.
Bergy bits	From 6 to 12 miles.
Growlers	From 1 to 3 miles.

Even in these conditions some growlers, of a size that could seriously damage the vessel, may not be detected at all.

Relying on radar and sonar

While radar and sonar, where fitted, are useful tools in looking for icebergs, they must not be relied upon.

DO

- Treat radar and sonar as aids that supplement the lookout, never replace it.
- Increase lookouts because of the known limitations of radar in detecting ice.
- Monitor the sea temperature where sonar is fitted, since temperature and salinity can prevent the signal reaching a nearby object.
- Study radar returns from different ice types in clear weather so they can be recognised in poor visibility.

DON'T

- Do not rely on radar in rough conditions when sea clutter extends beyond 1 mile, except for the detection of larger icebergs.
- Do not assume a clear screen means clear water; signals can be refracted or absorbed by the conditions.
- Do not expect a strong return from every berg; even the angle of the iceberg may provide a weak signal.
- Do not assume growlers will be seen; some that could seriously damage the vessel may not be detected.

Radar antenna temperature limits

The radar antenna working temperature range can normally be obtained from the specifications section of the radar manufacturer's manual.

-25 °C

Lower working temperature limit implied by IEC 60945

+55 °C

Upper working temperature limit implied by IEC 60945

Standards instead of values — Occasionally the specifications quote standards rather than actual temperature values. Any reference to IEC 60945 includes suitability for working temperature limits of -25 °C to +55 °C.

Verify scanner and motor — It should be verified that the working temperature range of the scanner and the scanner motor is suitable for use in sub-zero temperatures, and the range should be recorded.

Lubricants in the drive — Lubricants used in the scanner drive should be suitable for sub-zero temperatures.

SECTION 2.3

GPS, sonar, bridge windows and de-icing

Check the navigation equipment

Check that all other navigation equipment, such as GPS or DGPS, is functioning properly.

Sea temperature and sonar

If sonar is fitted, it is important to monitor the sea temperature, as this and the salinity can prevent the sonar signal reaching a nearby object.

Trim and ballast for ice

The ship will need to be in the right trim and ballast for ice navigation; it is important that the propellers are kept below the ice level.

Bridge windows and lights

Some method of avoiding freezing condensation on bridge windows will need to be used. Navigation light glasses can be smeared with Vaseline.

De-icing arrangements

A check should be made on all de-icing arrangements.

Trim and stability requirements in ice

Requirement	What it means
Draught	The draught must be kept between the load line and the ballast line during navigation in ice.
Propeller immersion	The ship must be ballasted and trimmed so that the propeller is completely submerged and is as deep as possible.
Tank filling	Tanks should be no more than 90 per cent full, to accommodate expansion if the contents freeze.
Trim	The vessel should have less than three feet of trim, to keep ice sliding under the vessel clear of the sea suction and propeller.
GM	The vessel must have an adequate GM for the conditions expected, including icing up and any icebreaking required.

Three sources of GM loss in ice

The vessel must have an adequate GM to allow for each of the following.

Slack water in tanks

The free surface effect of slack water in her ballast and fresh water tanks.

Icing up of the upper works

The loss of GM caused by ice accumulating on the upper works, high above the waterline.

Breaking ice floes

The virtual loss of GM experienced when breaking ice floes, which acts on the ship in the same way as grounding.

SECTION 2.6

Safety equipment checks

As part of the check on the ship's external systems, safety gear should be inspected.

Searchlights and illumination

Are the searchlights and illumination fully functioning?

De-icing compounds

Does the vessel have an adequate supply of one of the proprietary commercial de-icing compounds?

Protective clothing and lines

Will the crew have access to adequate protective clothing, harnesses and safety lines?

High visibility sections

Foul weather and high visibility clothing should be worn on all external operations during foul weather. Where the gear has no built-in high visibility sections, suitable gear such as armbands should be worn.

Harnesses, lines and protective clothing

DO

- Provide an approved safety harness to each person required to go out on deck.
- Make additional harnesses available on request if required.
- Ensure that any operation requiring a harness is carried out by a person wearing one, normally the most experienced person.
- Issue protective clothing and equipment to all those involved in work in extreme weather.
- Keep protective clothing comfortable and well maintained, and provide training in its use.

DON'T

- Under no circumstances carry out any type of work requiring a safety harness without the harness.
- Do not allow protective clothing that leads to an increase in other risks.
- Do not send people out in foul weather gear without high visibility sections or armbands.
- Do not accept that persons working in extreme weather feel anything less than totally safe at all times.

SECTION 08

Keeping a Keen Watch

08

A careful lookout must be maintained at maximum levels at all times, and the bridge crew must be kept on full alert.

SECTION 2.4

What a keen watch means in ice

As always, a proper lookout should be maintained, but if an ice zone is to be encountered then the risk of meeting obstacles is significantly raised.

A cautious business

Picking a safe passage through fields of ice, often shed by large icebergs, is a cautious business.

Know what you are looking at

The Officer of the Watch should be aware of the different types of ice floes and icebergs he is likely to encounter.

Familiar with identification

He should be familiar with the identification of ice, its type and formation, sea ice, glacier ice, icebergs, detection of pack ice, ice floes, other ice types and the relevant terms.

Constant vigilance on systems

Constant vigilance must be paid to all ship systems such as radar, sonar and radio. The bridge crew must be kept on full alert at all times.

SECTION 2.4

Bridge manning when conning in ice

- When navigating in ice the Chief Mate will usually assist the Master in conning the vessel from the bridge.
- One mate and an able seaman may be posted externally, on the bow, as lookouts to assist in the ice conning.
- One mate is required in the wheelhouse plotting continuous positions.
- Extra watches are to be set when encountering ice, and consideration made to putting a lookout forward.
- As navigating in ice may be a time-consuming process, optimum use should be made of the bridge team and all personnel on board.
- Rest and work hours requirements must still be observed.
- Consideration should be given to doubling up the bridge watchkeepers, with the Master or Chief Officer accompanied by the 2nd or 3rd Officer.

The watch gets tired

Officers on watch can be on the bridge for a long time. Even once the ice watch is secured, somebody has to stand watch, and that officer will already be tired from ice operations.

Means of detecting ice

Vessels should have means to detect ice. Such means may include the following.

Searchlights

Narrow beam searchlights, mounted on each bridge wing or at the bow.

Thermal imaging

Thermal imaging equipment able to distinguish ice from open water.

Ice radar

Ice radar, used with an understanding of its limitations.

Visual lookout forward

A visual lookout posted forward, which remains the most reliable means in many conditions.

SECTION 2.4

Searchlight specification

Requirement	Specification
Beam type	Searchlights should be of a narrow beam type.
Position	Mounted on each bridge wing or at the bow.
Height	Bridge wing lights should be below conning position eye level, to reduce disturbance during heavy snowfall.
Power	Minimum 2 kW halogen, or 1 kW xenon.
De-icing	Fitted with an adequate means of de-icing, to ensure proper directional movement.

SECTION 2.4.1

Why icebergs are hard to see

Icebergs can be difficult to see, so great caution is needed.

Concealed in fog

They can be concealed in fog, or blend in with the grey sky and sea.

Small above, large below

They can appear small above the surface but be very large beneath it.

Darkness and the polar winter

You may be under way at night time, or during the long Arctic winter.

Know where to expect them

British Meteorological Routing Charts are available to show the most likely ice zones and iceberg routes, and help decide when to set up observation routines as the zones are approached.

SECTION 2.4.1

Agencies that provide ice information

Many agencies provide updated information on ice and sea conditions and the proximity of icebergs. Their services are invaluable to mariners and very often free.

United States Coast Guard

Advice and current ice information, including the International Ice Patrol.

Canadian Coast Guard

Ice advisory services for Canadian waters.

Finnish Maritime Authority

Winter traffic information for the Baltic.

European Space Agency

Satellite-derived ice information.

Websites and radio

Advice is published on websites and broadcast by radio communications.

SECTION 2.4.1

How the authorities build the ice picture

Remote sensing

- These authorities gather information by satellite.
- Tracking aircraft cover the areas of greatest concern.
- Other sensors supplement the aerial and satellite coverage.

First-hand sightings from shipping

- They also rely on first-hand sightings from shipping.
- Whenever an iceberg is spotted, its position, approximate size, speed and course must be relayed to the relevant authority within the area.
- This maintains up-to-date databases and helps other ships.

Reporting an iceberg sighting

The authorities gather information by satellite, tracking aircraft and other sensors, but they also rely on first-hand sightings from shipping.

1. Sight it

The lookout or the OOW sights the iceberg and informs the Master.

2. Fix it

Establish its position, together with the time and date of the initial sighting.

3. Describe it

Record its approximate size, speed and course.

4. Report it

Relay the report to the relevant authority within the area, which maintains up-to-date databases and helps other ships.

Arctic iceberg drift — the latitudes

In the Arctic many icebergs originate in the glaciers of Greenland and drift south with the prevailing currents.

Feature	Detail
East Greenland current	Carries bergs south beyond Cape Farvel on the 60th parallel, towards Newfoundland and the east coast of North America.
Baffin Bay glaciers	Many more bergs come from the glaciers of Baffin Bay; many run aground locally and go no further.
Canadian and Labrador currents	Significant numbers drift slowly south to threaten shipping lanes south of the 48th parallel off Newfoundland.
Southern limit	Usually they float no further south than the 42nd parallel.

SECTION 2.4.1

Iceberg numbers in the north-west Atlantic

40,000

Estimated icebergs afloat in Baffin Bay at any one time

200+

Average number each year reaching lanes south of the 48th parallel

900

Peak number that has been reached in a single year

The numbers vary — The numbers vary each year, so the previous season is no guide to the present one.

Grounded bergs — Many bergs run aground locally in Baffin Bay and go no further south.

SECTION 2.4.1

The Antarctic ice zone

The published zone

- The Antarctic ice continent is much larger and deeper than the ice cap of the North Pole.
- The ice zone around it spreads north to between the 58th and 62nd parallels.
- Icebergs are possible anywhere within this area, especially in the glacial areas of the Weddell and Ross Seas.

Beyond the zone

- It is not uncommon to detect bergs well north of these latitudes.
- They have been located north of the 40th parallel in the South Atlantic.
- The published limit is a planning aid, not a boundary you can rely on.

SECTION 2.4.1

Listening for ice

An alert lookout is not restricted to using your eyes.

Breaking icebergs

Listening for the sounds of breaking icebergs is important.

Waves breaking over

So is the detection of waves breaking over them.

Absence of sea

The absence of sea in a fresh breeze could indicate a large object to windward.

Thunder without a storm

Thunderous sounds but no storm could suggest an iceberg breaking up.

Growlers in the water

Growlers and small pieces of ice could indicate a disintegrating iceberg.

SECTION 2.4.1

Signs that pack ice is close

Pack ice is easier to look out for, but it can still be difficult to see in certain conditions. Pale sun, fog and mist can create ice blink.

Ice blink in blue skies
With blue skies, ice blink can appear as a luminous yellow haze on the horizon in the direction of the ice.

Ice blink under overcast
With an overcast sky it can appear as a white glare on the clouds.

The sea smooths abruptly
Sure signs of the approach of an ice pack are the abrupt smoothing of the sea and the gradual lessening of the ordinary ocean swell.

Small floes ahead
Small ice floes appearing in the water are a direct indication of the pack ahead.

Wildlife
Certain types of wildlife, including seals, walruses and birds, also indicate the approach of the pack.

SECTION 2.4

Lookout discipline in an ice zone

DO

- Maintain a careful lookout at maximum levels at all times.
- Post a lookout forward, and consider a mate and an able seaman on the bow to assist in ice conning.
- Keep the bridge crew on full alert, monitoring radar, sonar and radio.
- Reduce speed without hesitation if an iceberg is sighted without warning, or if signs warn that one may be close.
- On a dark night at slow speed ahead, treat the sound of breakers where no land is expected as a large object to be avoided.

DON'T

- Do not treat radar or sonar as a substitute for a visual lookout.
- Do not reduce lookout levels because the screen looks clear.
- Do not wait for confirmation before reducing speed on a sighting.
- Do not leave the ice watch to a single tired officer; double up the watchkeepers.

SECTION 09

Manoeuvring in Ice

It is vitally important to maintain freedom of manoeuvre. Keep moving, work with the ice not against it, and remember that excessive speed leads to ship damage.

SECTION 3.1

When ice is confirmed

As a general rule, ice is an obstacle. When ice is confirmed, a fixed sequence of actions follows.

The Master to the bridge

When ice is confirmed the Master must be informed, and his presence on the bridge is now necessary.

Report to the authorities

He must make a report to the local authorities. This also applies if the vessel is in an ice zone and an iceberg is sighted.

Advise the Chief Engineer

The Master must advise the Chief Engineer that they are approaching ice, as the vessel's speed will have to be reduced.

Continuous plotting

One mate is required in the wheelhouse plotting continuous positions.

What the Master reports to the authorities

Item	Detail reported
Type of ice	The type of ice encountered or the iceberg sighted.
Position	The position of the ice, and the exact position of the ship.
Time and date	The time and date of the initial sighting.
Temperatures	Air and sea temperature, if below freezing.
Wind	The direction and force of the wind.
Accumulation	Ice accumulation on the ship.

SECTION 3.1

The cold the crew must be equipped for

The crew must have the appropriate working and survival clothing for the conditions.

-20 °C

Ambient temperatures that can be experienced

-30 °C

Equivalent temperature with wind chill

A poorly equipped sailor — A poorly equipped sailor will not last long and will not be able to carry out his duties or survive emergency or abandon ship stations.

Monitor the build-up — Ice build-up on the ship's superstructure must be monitored at all times, to ensure it does not become excessive.

Waves and spray — In gale or storm conditions where ambient temperatures are below zero, waves and spray can build up and severely affect the stability of the ship.

SECTION 3.2

Conditions change faster than the plan

Circumstances can change rapidly in ice zones. Temperatures can suddenly drop, and water that is free one minute can very quickly become solid.

Maximum vigilance

- Maximum vigilance is needed from every officer and crew member.
- Conditions and the ship's behaviour must be monitored constantly.

The watch will tire

- Officers on watch can be on the bridge for a long time.
- Even once the ice watch is secured, somebody has to stand watch, and that officer will become tired from ice operations.

Double up the watch

- Consideration should be given to doubling up the bridge watchkeepers.
- The Master or Chief Officer is accompanied by the 2nd or 3rd Officer.

Choosing the point of entry

Never underestimate the hardness of ice. It can vary in thickness, the floes can be of different sizes and therefore strengths, and it can also be moving in a current.

DO

- Use the ice edge as the point of entry where the ice has a definite beginning.
- Always try to enter at 90 degrees to the edge.
- Enter on the leeward side where possible.
- Find an alternative entry point if ridging or hummocking is severe.
- Keep the Chief Engineer informed of the moment of entry, so that he can respond to demands for extra power immediately.

DON'T

- Do not enter through severe ridging or hummocking; ridged ice may be far too deep and could severely damage the hull.
- Do not enter on the windward edge, which will be more compact with greater wave action.
- Do not enter at an oblique angle to the ice edge.
- Do not enter at anything other than reduced speed.

SECTION 3.3

The three ship-handling rules

It is vitally important to maintain freedom of manoeuvre. The three basic ship-handling rules are as follows.

Keep moving

Keep moving, even if very slowly. Freedom of manoeuvre is lost as soon as the ship stops.

Work with the ice

Try to work with the ice, not against it. The ice will not give way to the ship.

Speed causes damage

Excessive speed leads to ship damage. In ice zones, lookouts and radar operators must reduce speed without hesitation if an iceberg is sighted without warning.

SECTION 3.4

Communication with the engine room

Keep the engine room informed

- Regular communications with the engine room should be maintained at all times.
- The Chief Engineer and his staff must be aware of the prevailing conditions.
- They must be ready to increase and reduce power when necessary.

Sea chests at optimum efficiency

- The Chief Engineer must ensure that the cooling water sea chests are working at optimum efficiency.
- Cooling water intakes may become frozen, and sea water strainers and filters must be kept clean.

Entering the ice

1

Enter at slow speed

Entry into the ice must be at slow speed, after the Master has been notified of the approach to the ice edge.

2

Increase slowly

Depending on conditions, speed should then be increased slowly to maintain headway and control.

3

Watch for hard ice

In an area of light drift ice, lookouts should watch for large floes or fragments of old hard ice that pose significant threats. These must be avoided.

4

Searchlights and helmsman

At night in these conditions all searchlights must be used, with the helmsman in place ready to take evasive action.

Meeting a large floe

If possible avoid large floes, as they may have underwater spurs. Where avoidance is impossible, the handling sequence is fixed.

Avoid it if you can
Large floes may have underwater spurs that will reach the hull, the propeller and the rudder.

Push it clear
If avoidance is impossible, it may be possible to push the floe out of the way.

Reduce power as it swings
Once it starts to swing to one side, reduce power and allow it to pass clear.

Hit it squarely
If collision is unavoidable, hit the ice as squarely as possible.

Never a glancing blow
A glancing blow could damage bow plates and swing the ship, bringing the stern onto the floe and damaging the propeller and rudder.

SECTION 3.6

Setting speed in ice

The best speed to maintain depends on two factors, and it must be adjusted as conditions change.

Factor or situation	What it requires
The vessel’s tonnage	A heavier ship carries more way into the ice and needs a correspondingly lower speed.
The density of the ice	The best speed to maintain depends directly on how close the ice is.
Varying concentration	If concentrations of ice vary, so must the ship’s speed.
Faster speeds in light ice	Faster speeds in light ice could mean striking heavier ice with greater way on the ship, and cause damage.
Sudden heavy ice	If a sudden heavy section of ice is encountered and stops the ship, the engines must be prepared to go full astern at any time.

Use of engines and rudder

Going astern must be done with great caution, as the propeller and the rudder are the most vulnerable parts of the ship.

DO

- Put the rudder amidships with the engines turning slowly ahead before going astern; this washes the ice astern clear.
- Have crewmen confirm that the stern is clear before the ship comes astern.
- Use frequent rudder to the hard over positions where the intention is to slow the vessel without loss of steerage way.
- Keep the engines prepared to go full astern at any time.

DON'T

- Do not make violent rudder movements except in an emergency; they may swing the stern into the ice.
- Do not use so much rudder that the vessel is brought to a complete stop, which is very dangerous in freezing conditions.
- Do not go astern before the stern has been confirmed clear.
- Do not go astern without first washing the ice clear of the stern.

SECTION 3.7

Ramming and backing

Ramming the ice is very dangerous and acceptable only if the ship is in danger, and forcing a passage through to open water or less heavy ice is the only alternative.

A last resort only

It requires great caution, and you need to be certain that the bow will crack the ice rather than the ice damage the ship.

The method

Ram the ice to break it by sheer impact and weight, then reverse and try again. The action is repeated until access to clear water is made.

When to stop

If progress is slight and the channel created is not considerably wider than the beam, the procedure should be stopped.

Damage can be fatal

Extensive damage could prove fatal even if the ship reached clear water. Avoid getting trapped in the ice.

When ramming is the better option

If trouble cannot be avoided, it is often better to ram the ice and embed the bow than collide with another ship or the icebreaker. Ramming is also the best option if there is any danger of the propeller and rudder hitting the ice.

Freeing a trapped ship

When a ship becomes trapped an icebreaker will usually be required. However, there are a few methods the ship can try to free itself.

Full ahead and full astern

If the propellers are not completely embedded, the ship can try going full ahead and then full astern.

Alternate full helm

Alternating full helm in both directions may swing the ship and loosen the ice grip to allow movement ahead.

Shift ballast

Shifting the ballast from side to side may also help to break the grip of the ice.

Chains and winches

The ship's anchor chains and winches can be used to assist in working the ship free.

Working with an icebreaker

The icebreaker commands

When working with an icebreaker, the Master of the icebreaker is in overall command and directs all operations.

Know the signals

All officers on the bridge must be thoroughly familiar with icebreaker signals as shown in the International Code of Signals.

Execute immediately

All instructions from the icebreaker must be acknowledged and executed immediately.

Watch and listen continuously

The ship must continuously watch and listen for the icebreaker's signals, and for those of other ships that may be being assisted at the same time.

Monitor the VHF channel

The agreed VHF channel must be continuously monitored.

Convoy speeds and distances

The escorted vessel must follow the path cleared by the icebreaker, and it is important to maintain the same speed as the icebreaker.

6 – 7 kn

Typical icebreaker speed in open ice

5 kn

Maximum speed in close ice

Thicker ice, lower speed — Speed will be less than 6 to 7 knots in thicker concentrations, and no more than 5 knots in close ice.

Minimum distance — In convoy the icebreaker Master issues instructions on the order of ships and the minimum distance between them. The minimum distance is that required for an emergency stop with the engines full astern.

When the distance is reduced — The distance may be less if the pressure in the ice pack is so great that the channel will not remain open long enough for the ship to pass.

Convoy discipline

- The escorted vessel must follow the path cleared by the icebreaker.
- It is often better to travel in convoy, and in this case the icebreaker Master issues instructions on the order of ships.
- The icebreaker may need to hit the ice at speed to crack it and force a path.
- The vessels following must then proceed without delay, before the gap closes again.
- All signals must be immediately and precisely repeated to the ship behind, and passed down the line.
- If in a channel made by an icebreaker, follow strictly the pilot’s or icebreaker’s advice and commands.

The line depends on you

A signal that is not passed on stops at your ship. Every vessel in the convoy repeats each signal immediately and precisely to the ship astern of her.

SECTION 3.11

Emergencies — towing and damage

The icebreaker decides
The icebreaker decides when a ship needs to be taken into tow.

Towing gear rigged
The ship must be prepared, with towing gear rigged at all times, in case towing is necessary.

Engines on orders only
A ship being towed by an icebreaker must use its engines only in accordance with the orders given by the icebreaker.

Report damage immediately
Any damage or suspected damage must be reported to the icebreaker immediately.

Searchlight off when stopped
If the assisted vessel stops because of ice conditions and the searchlight has been in use, it must be switched off as long as the ship is stationary.

SECTION 3.11

Fork towing and the last resorts

Fork towing

- If ice conditions deteriorate during icebreaker assistance, towing might be the only safe and prudent way to continue.
- Towing should be done in the normal manner, using so-called fork towing.
- The vessel’s bow is taken inside the towing fork and two cables from the icebreaker are attached to the assisted vessel’s bitts, which are designed to withstand the stresses of towing.

If trouble cannot be avoided

- It is often better to ram the ice and embed the bow than collide with another ship or the icebreaker.
- Ramming the ice is also the best option if there is any danger of the propeller and rudder hitting the ice.
- When a ship is trapped, the engine should be kept running and the propeller turning, if possible, to prevent ice forming at the stern.

SECTION 10

Reports, Training and the Polar Code

Where the ice reports come from, the regional publications that must be carried, and the training, risk assessment and Polar Code requirements that govern the operation.

10

SECTION 2.5

Ice condition and meteorological reports

An essential part of the briefing is the meteorological information.

Conditions and forecasts

All the officers on board must be aware not only of the types of ice conditions to be experienced, but also of the prevailing weather forecasts.

National agencies

Countries that depend on maintaining traffic lanes during ice conditions have agencies that provide detailed meteorological reports, together with substantial information on ice conditions based on monitoring and analysis.

Contact the authority

The Master can make contact with such authorities at any time he is approaching national coastal state waters.

Verify what you receive

The Master should conduct verification of hydrographic, meteorological and navigational information from all sources available, to ensure the information is correct and correctly addressed.

Two key agencies and the receiving channels

Canadian Coast Guard

www.ccg-qcc.ca — detailed ice and meteorological reports for Canadian waters.

Finnish Maritime Administration

www.fma.fi — winter traffic assistance and ice information for the Baltic.

Direct internet access

All fleet vessels have direct internet access, so ice reports should be downloaded from the internet.

NAVTEX, Inmarsat C, Weather Fax

NAVTEX, Inmarsat C and Weather Fax can also be used for receiving reports.

Where to download ice reports

The following links are provided for use on board.

Area	Source
Baltic Sea	http://www.smhi.se/oceanografi/istjanst/havsis_en.php
Canadian waters	http://www.ccg-gcc.gc.ca/Icebreaking/Ice-Navigation-Canadian-Waters/Navigation-in-ice-covered-waters
Canadian Coast Guard	www.ccg-qcc.ca
Finnish Maritime Administration	www.fma.fi

Ensure the vessel's NAVTEX has the ice report receiving function activated.

SECTION 2.5.1

Ice Navigation in Canadian Waters

Every ship of 100 tons gross tonnage or over navigating in Canadian waters in which ice may be encountered is required to carry, and make proper navigational use of, the Canadian Coast Guard publication Ice Navigation in Canadian Waters. The document is in two parts.

Part I — Operating in Ice

- Pertains to operational considerations.
- Communications and reporting.
- Advisories.
- Icebreaker support.

Part II — Additional Information

- Educational in nature.
- Familiarises watchkeepers with the Canadian ice environment.
- Navigation procedures in ice.
- Vessel performance in ice.

Carriage, editions and the Finnish authority

Carriage requirement

Every ship of 100 tons gross tonnage or over navigating in Canadian waters where ice may be encountered must carry and make proper navigational use of the publication.

The current edition

The 1999 version of Ice Navigation in Canadian Waters is the most current version and there have been no amendments. As the information does not vary from year to year, the document does not require frequent revisions.

Amendments by mailing list

Canadian Hydrographic Services, as distributor, maintains a mailing list for those who purchase the publication, including those on the list of their previous distributor. Any amendments are sent out automatically whenever required.

SECTION 2.5.2

The Finnish Maritime Administration

The Finnish Maritime Administration is the authority responsible for the following, and ensures that the basic operational conditions for merchant shipping and sea transport are maintained and continually improved.

Maritime safety

Overall responsibility for maritime safety in Finnish waters.

Winter traffic assistance

Assistance to traffic during the ice season.

Fairway maintenance

Maintenance of the fairways used by commercial shipping.

VTS and pilotage

Vessel traffic services and pilotage provision.

Charting and ferries

Hydrographic charting, and provision of ferry services to the archipelago communities.

Training, risk assessment and the ice adviser

Use every opportunity to train

All possibilities should be used to train and improve the ice navigation knowledge of ship and office staff.

The Training Manual

The detailed training plan and curriculum of ice training is given in the Company's Training Manual, which also sets out the company requirements for familiarisation to ice navigation.

Risk assessment is essential

According to company policies all ships should assess risks prior to ice navigation. Due to the natural risks and dangers of ice navigation, it is essential to make a risk assessment.

The ice adviser

All actions to make a safe passage in ice, and to gain experience and knowledge of ice navigation, must be taken by the Master. Under these circumstances the Master can demand an ice adviser, by approval of the Company.

Risk assessment before ice navigation

Risk assessment procedures are described in the Company Management Manual.

1. Assess on board

The ship assesses the risks of the intended ice navigation, covering precautions, preparations and necessary controls.

2. Send to the office

The completed risk assessment form must be sent to the office before commencement of ice navigation.

3. Office review

Office staff review the assessment against the intended operation.

4. Additional instructions

Where there are any, additional instructions and assessments will be sent back to the ship by office staff.

SECTION 4.4

Polar Code — navigational equipment functionality

For navigation in polar waters, vessels should comply with Polar Code requirements and be certified accordingly by the classification society to ensure safety of navigation.

Receive ice information

Ships shall have the ability to receive up-to-date information, including ice information, for safe navigation.

Retain functionality

Navigational equipment and systems shall be designed, constructed and installed to retain their functionality under the expected environmental conditions in the area of operation.

Heading and position

Systems for providing reference headings and position fixing shall be suitable for the intended areas.

Detect ice in darkness

Ships shall have the ability to visually detect ice when operating in darkness.

Indicate when stopped

Ships involved in operations with an icebreaker escort shall have suitable means to indicate when the ship is stopped.

Voyage planning under the Polar Code — the route

The voyage plan shall take into account the potential hazards of the intended voyage. The Master shall consider a route through polar waters taking the following into account.

- The procedures required by the Polar Water Operational Manual (PWOM).
- Any limitations of the hydrographic information and aids to navigation available.
- Current information on the extent and type of ice and icebergs in the vicinity of the intended route.
- Statistical information on ice and temperatures from former years.
- Places of refuge along the intended route.

Displaying the information

To comply with the functional requirement of the Polar Code, ships shall have means of receiving and displaying current information on ice conditions in the area of operation.

Voyage planning under the Polar Code — the environment

The remaining considerations concern the environment along the route and the assistance available.

Marine mammals

Current information and measures to be taken when marine mammals are encountered, relating to known areas with densities of marine mammals, including seasonal migration areas.

Routeing and VTS

Current information on relevant ships' routeing systems, speed recommendations and vessel traffic services relating to those same areas.

Protected areas

National and international designated protected areas along the route.

Remote from SAR

Operation in areas remote from search and rescue (SAR) capabilities.

SECTION 4.5

Communication characteristics in high latitudes

Radio communications beyond line-of-sight limits in the Arctic pose certain problems that occur only rarely at lower latitudes.

Ionospheric disturbances

These problems are mainly caused by ionospheric disturbances which affect the behaviour of radio waves in the LF, MF, HF and VHF bands.

The transmitter range

The standard shipboard transmitter required for high-latitude operations usually covers a frequency range down to 175 to 195 kHz, depending on the model.

Very little radiated power

At the lower end of the range the radiated power obtained may be very small, of the order of 0.6 to 1.5 per cent of the rated output power.

Why the power is lost

This is caused by the inability to load the antenna properly because of its short effective length. Radiation is further decreased by the necessarily short vertical section of the antenna, upon which radiation depends.

Transmitter and antenna figures

Parameter	Figure and effect
Transmitter lower limit	175 to 195 kHz, depending on the model of the standard shipboard transmitter.
Radiated power at that limit	0.6 to 1.5 per cent of the rated output power.
Optimum radiation	Usually realised at around 500 kHz with current shipboard antennas; effective radiation decreases as the operating frequency is decreased.
LF operation at 1 to 2 kW	Using a standard shipboard antenna at the lower range produces severe arc-overs and large losses in antenna trunks.
Cause of the losses	High standing-wave ratios, with high voltage occurring at the insulators, caused by inadequate effective lengths.

SECTION 4.5

Improvising for the lower frequencies

For these important lower frequencies to be used during periods of ionospheric disturbance, arrangements may have to be made on board.

Temporary antennas

Install temporary transmitting antennas of the greatest length and height practicable.

Parallel existing antennas

The possibility of paralleling existing antennas in series might be considered.

Balloon suspension

Emergency use of balloon suspension should not be overlooked.

Plan the communications

Ice passage plans place a greater emphasis on maintaining communication than open-water plans. In high latitudes that emphasis has to allow for the physical limits of the equipment on board.

Ice gives no second chance

Ice is an obstacle, it is very strong, and it commands great respect. Prepare the ship, brief the team, keep a keen watch and keep moving — slowly.

- Company approval before any ice entry
- Complete Form BCL-11 and the risk assessment
- Extra watches and a lookout forward
- Never force ice, never exceed 5 knots in pack